Enerlytics AI Platform

Course Name: GenAI

Institution Name: Medicaps University – Datagami Skill Based Course

Sr no	Student Name	Enrolment Number
1	Srashti Bhadoriya	EN22CS301982
2	Pranjal Sharma	EN22CS301731
3	Roshni Patidar	EN22CS301825
4	Riya Patidar	EN22CS301813
5	Ritik Patidar	EN22CS301809

Group Name: 10D7

Project Number GAI-33

Industry Mentor Name: Aashruti Shah

University Mentor Name: Dr. Hemlata Patel

Academic Year: 2025-2026

Problem Statement

The energy industry operates on massive volumes of technical documentation, ranging from grid compliance reports to renewable energy manuals. However, the sector faces several critical bottlenecks that prevent efficient information leverage:

- **Fragmented Information Silos**: Critical technical data is often trapped in unstructured PDF formats and legacy document types across isolated repositories. Engineers spend an estimated 30-40% of their daily bandwidth simply searching for relevant technical specifications across thousands of pages.
- **Semantic Retrieval Blindness**: Conventional search methodologies (literal keyword matching or fail to understand conceptual relationships. For example, a search for "Grid Resilience" might fail to retrieve relevant sections on "Substation Hardening" if the exact keywords do not match, leading to critical specialized knowledge being overlooked.
- **The "Hallucination" Safety Gap**: General-purpose LLMs lack domain-specific grounding. In energy operations, an AI producing "hallucinations" (technically incorrect but confident-sounding advice) can lead to catastrophic financial losses or physical safety hazards in grid management.
- **Expert Knowledge Attrition**: As senior energy experts and grid operators retire, their tacit knowledge is lost. New operators require 24/7 technical assistance that current static manuals cannot provide, creating a "Knowledge Gap" in mission-critical environments.
- **Data Sovereignty & Critical Infrastructure Security**: Public AI platforms raise significant privacy concerns. Sending sensitive energy grid data to external public AI centers poses a national security risk, requiring a solution that can integrate secure, controlled data flows.
- **Rapid Regulatory Flux**: Energy policies and grid protocols change rapidly. Manually tracking updates across multiple regulatory bodies is humanly impossible, leading to compliance risks and operational delays.

There is a critical need for **Domain-Grounded Intelligence**—a system that is fast, secure, and produces verifiable results with a direct audit trail to its sources.

Project Objectives

- **Semantic Precision**: Implement high-accuracy vector retrieval to ensure AI responses are contextually relevant to energy engineering.
- **Ultra-Low Latency**: Leverage specialized hardware (Groq LPU) to provide real-time streaming answers.
- **Source Auditing**: Create a 1:1 mapping between AI responses and the original document source pages.
- **Zero-Downtime Architecture**: Implement automated API rotation to handle heavy user loads without service interruption.
- **End-to-End Security**: Ensure encrypted data flows and isolated user workspaces.

Scope of the Project

The scope includes the development of:

- A responsive React-based **Frontend Discovery Portal**.
- A high-performance **FastAPI Backend Orchestrator**.
- A **RAG (Retrieval-Augmented Generation) Pipeline** using local embeddings and cloud vector search.
- An **AI-Driven Gating Engine** to prevent vector database contamination.

Proposed Solution

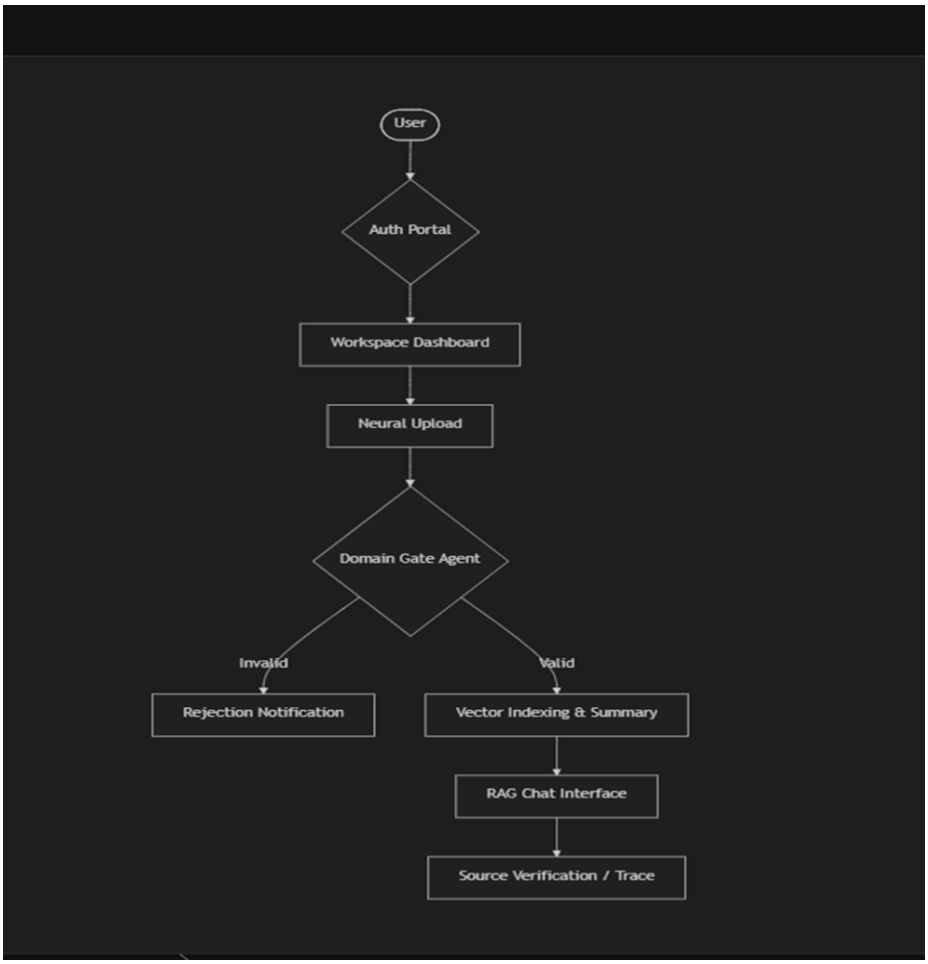
2.1 Key Features

- **3D Neural Discovery**: Interactive dashboard for workspace management.
- **Real-Time Knowledge Ingestion**: Synchronous PDF parsing and vectorization.
- **Semantic Domain Gating**: Automated verification of document relevancy.
- **Analytical Trace (The Evidence Engine)**: Visual confirmation of AI sources.
- **Asynchronous Chat Streaming**: Token-by-token generation for superior UX.

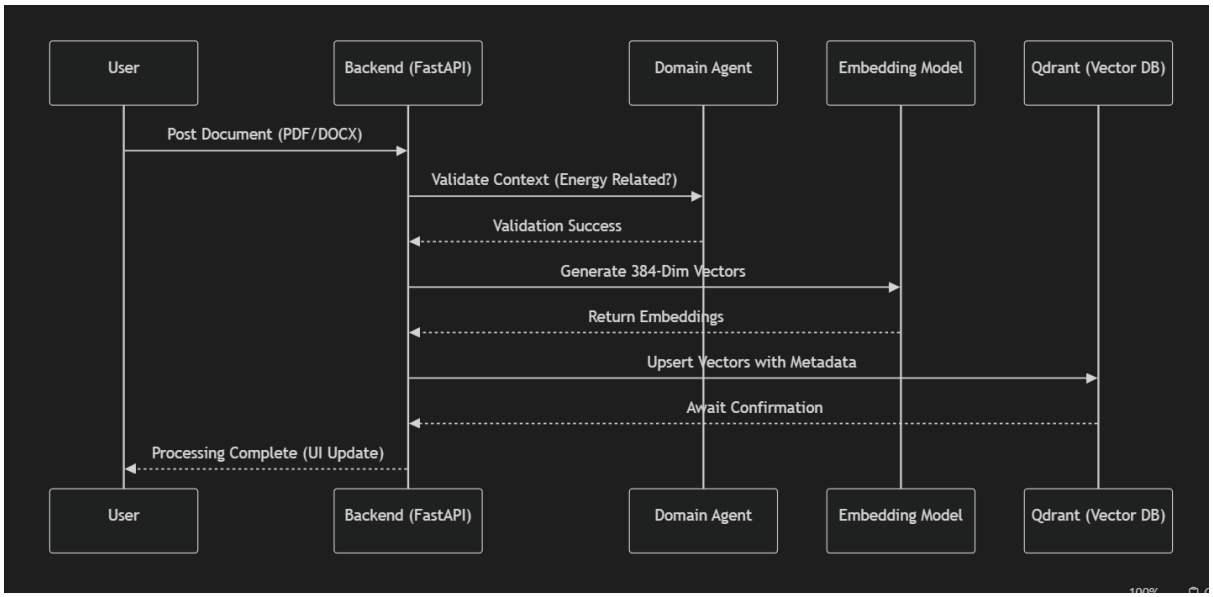
2.2 Overall Architecture / Workflow

The system follows a modular micro-service architecture designed for scalability.

A. Project Workflow Flowchart:



B. Document Ingestion Sequence Diagram:



2.3 Tools & Technologies Used

- **Frontend**: React 18 (Hooks/Context API), Vite (Bundler), Vanilla CSS, Framer Motion, Vanta.js.
- **Backend Architecture**: FastAPI (ASGI), Python 3.11, Pydantic (Schema validation).
- **Cognitive Engine**: Groq LPU (Llama 3.1 8B/70B), SentenceTransformers (all-MiniLM-L6-v2).
- **Data Persistence**: Qdrant Cloud (Vector Store), Redis Cloud (Chat Cache & History).

3. Results & Output

3.1 Methodology & Implementation

We implemented a **Sequential RAG Strategy**:

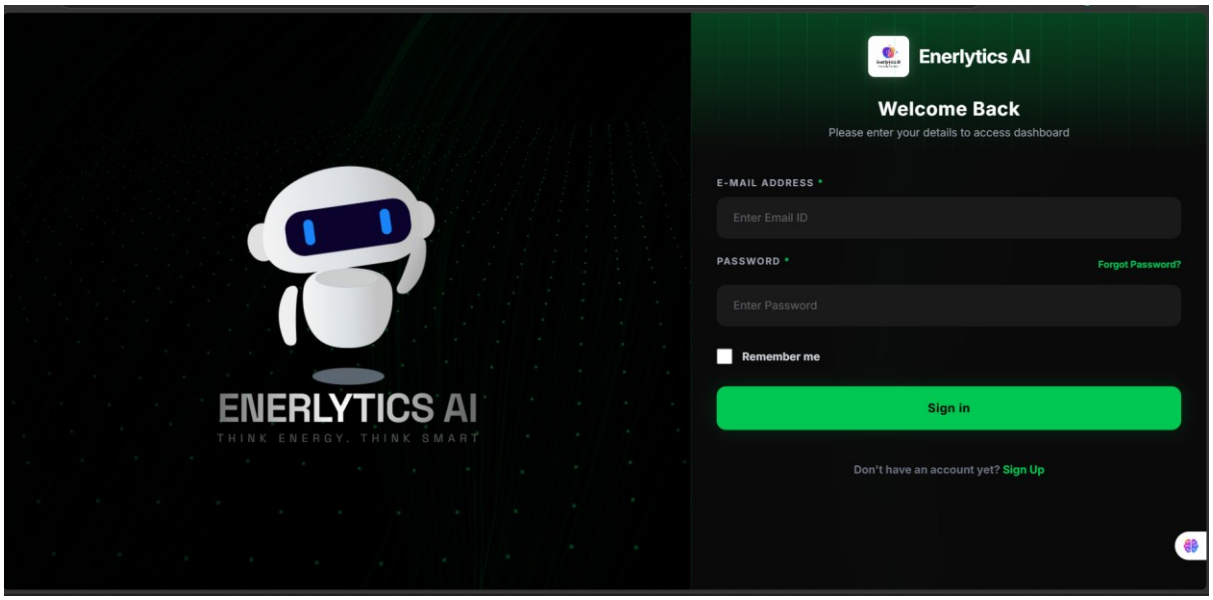
1. **Chunking**: Documents were split into 400-character segments with a 50-character overlap to maintain semantic continuity.

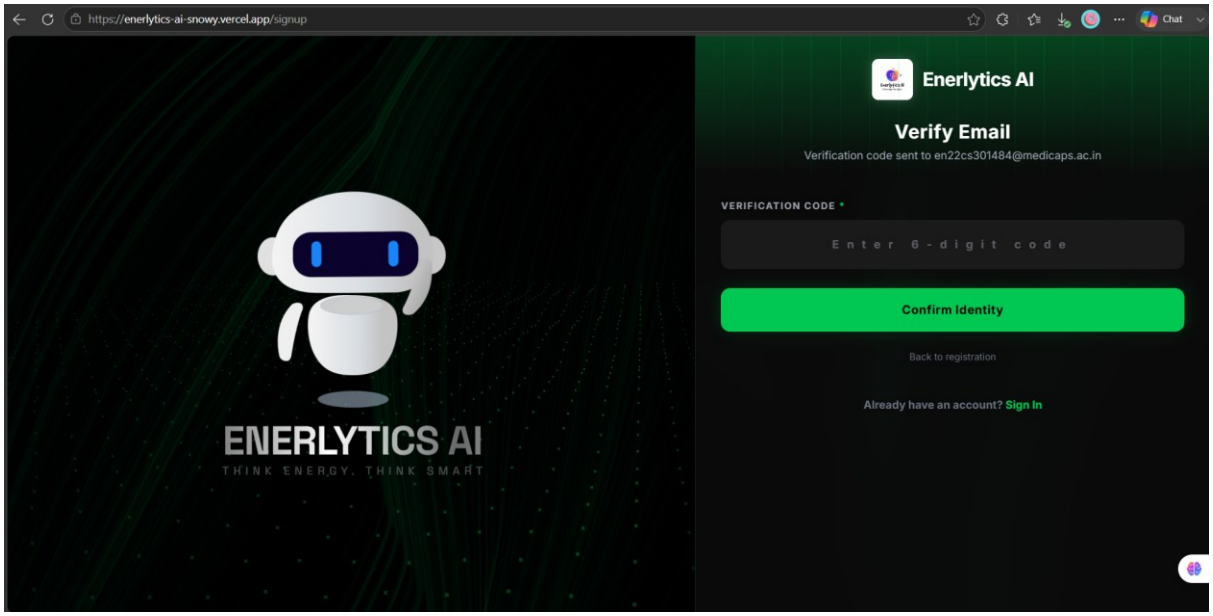
2. **Indexing**: Local CPU-optimized models were used for embedding to reduce costs.
3. **Retrieval**: Utilized **Cosine Similarity** with metadata filters (user_id) to fetch the top 6 relevant context chunks.
4. **Generation**: Integrated **Groq-Llama-3.1** via a secure API Rotator singleton.

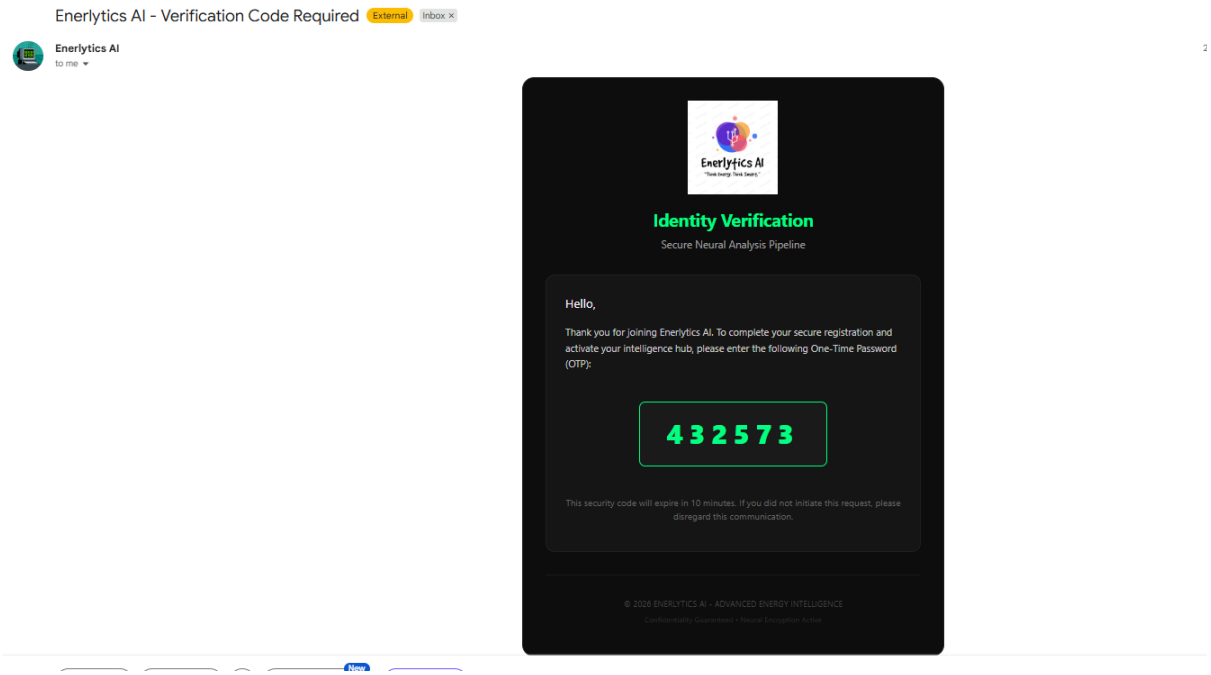
3.2 Screenshots / Outputs

To demonstrate the functional capabilities of the Enerlytics AI platform, the following interface snapshots are included:

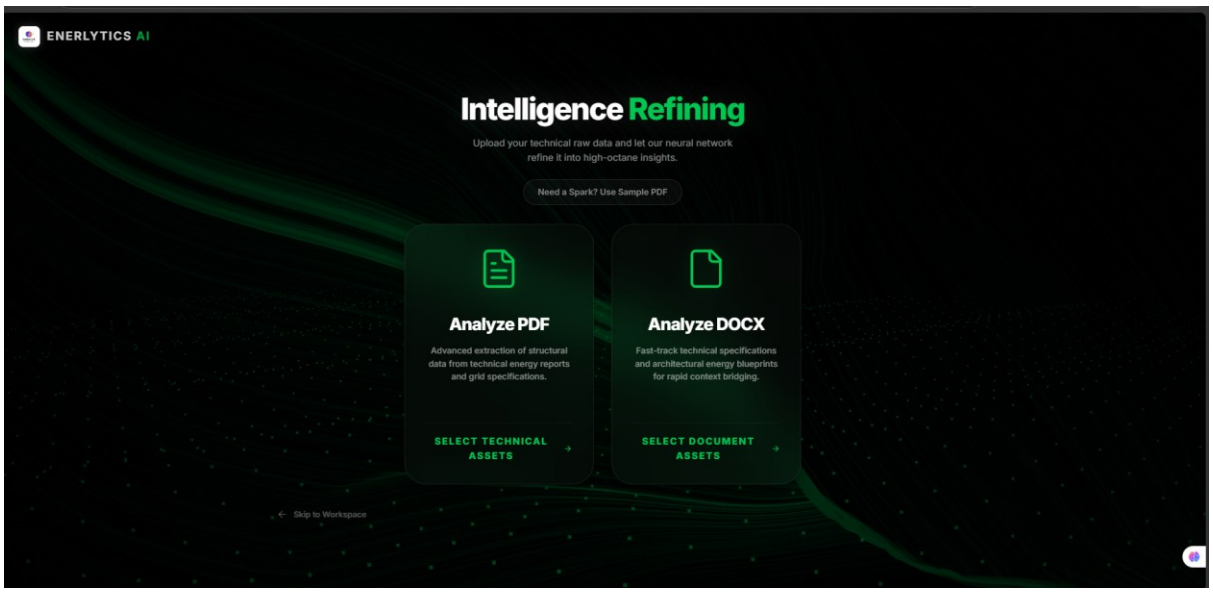
- **A. Neural Landing & Dashboard**: Showcases the 3D interactive interface and session management.
- **B. Secure Authentication (OTP)**: Demonstrates the multi-factor authentication flow.

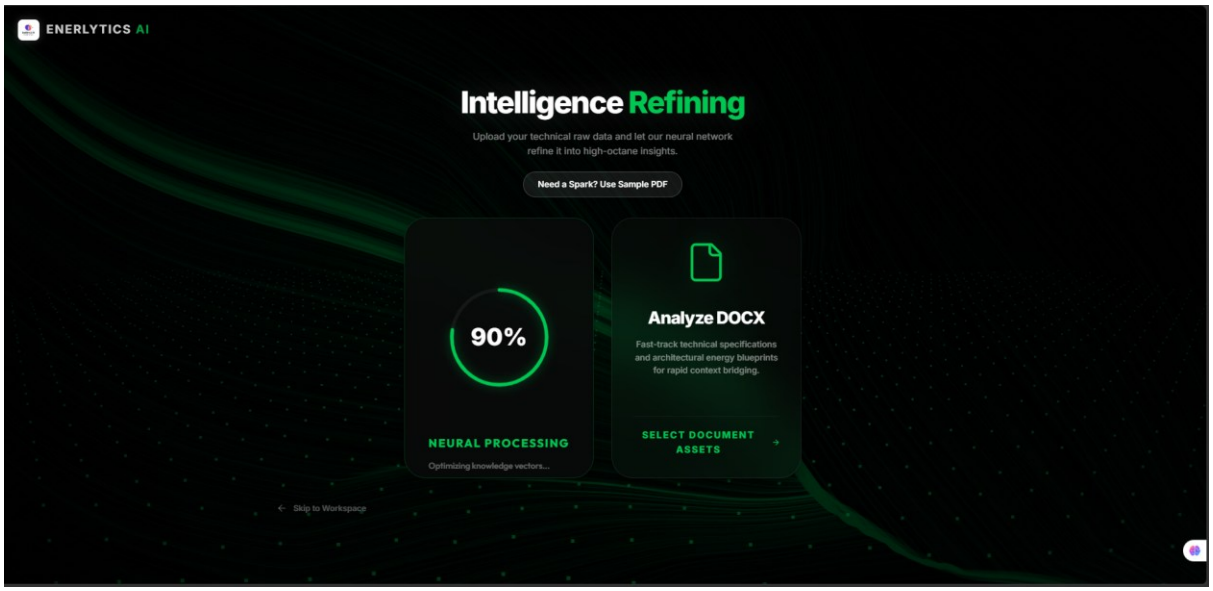




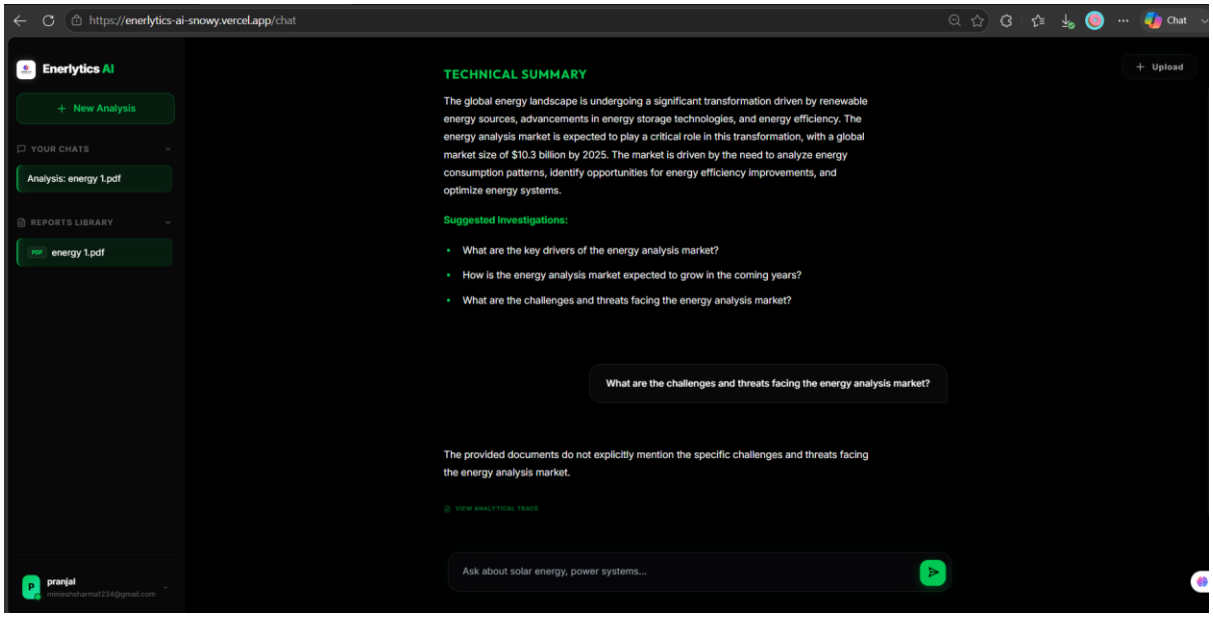


C. Document Ingestion & Gating: Visual proof of the AI Analysis Agent validating document context.

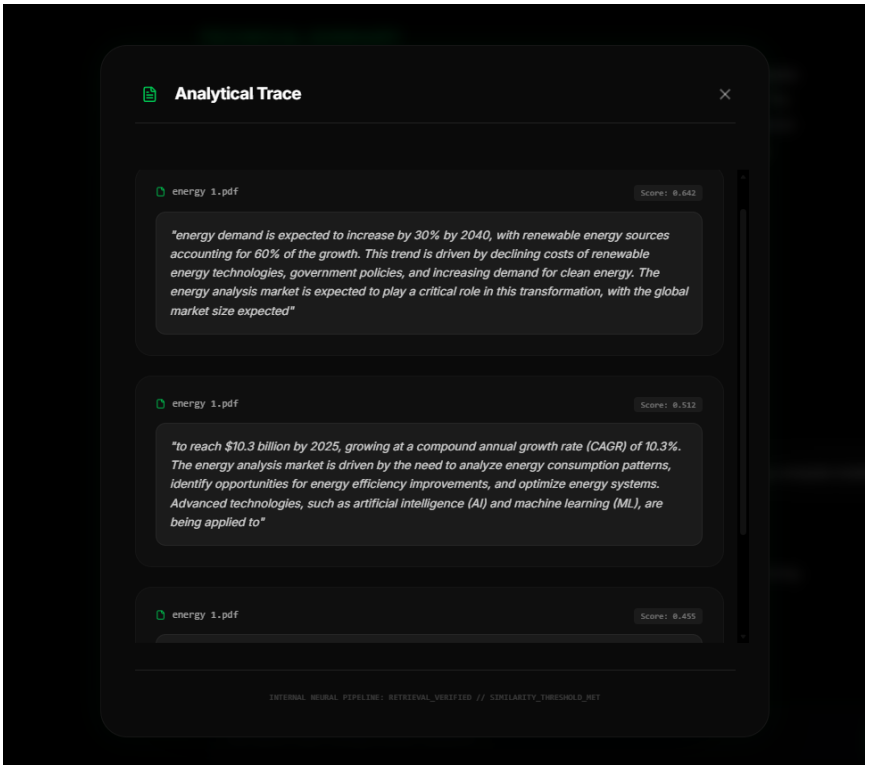




D. Cognitive Chat streaming: Real-time analytical conversation with the RAG engine.



E. Analytical Trace (Evidence Engine): Shows the 1:1 audit trail and document source highlighting.



5. Conclusion

Enerlytics AI represents a robust integration of modern Generative AI with specific industry requirements. The project successfully navigated technical constraints like memory management and rate limiting while delivering a premium user experience. **Key Learning & Growth**:

- Mastering **Asynchronous Python** for real-time applications.
- Understanding the deep mechanics of **Vector Embeddings** and semantic search.
- Developing secure, session-isolated web architectures using JWT and local cache.

6. Future Scope & Enhancements

- **Multi-Modal Analysis**: Allowing users to upload schematics and electrical diagrams for visual AI analysis.
- **Enterprise Collaboration**: Features for team-based shared knowledge bases.
- **Enhanced Re-ranking**: Implementing Cross-Encoders to further improve the relevance of retrieved chunks.

Refining Chat Interface Styling